Justin Mascarenhas

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Summary

Edge AI engineer who turns research models into real-time systems on embedded hardware: compressing, quantizing and accelerating networks until they fit a device's latency and memory budget, then building them into robots that act on what they see. M.S. Computer Engineering, RIT (2026).

Skills

Edge AI & Optimization: TensorRT, 8-bit quantization (PTQ & QAT), CUDA C++, Memory coalescing, pynvml profiling, ONNX / TFLite export
ML & Models: PyTorch, Hugging Face Transformers, YOLO, Transformers from scratch, ASR (faster-whisper), OpenCV
Platforms & HPC: NVIDIA Jetson, Raspberry Pi, Arduino, MPI, SLURM, Linux / systemd
Languages: Python, C/C++, CUDA, Cython, Bash

Experience

AI/ML Intern | **Indian Institute of Technology Bombay** Jan 2024 – Mar 2024

- Fine-tuned DONUT, an OCR-free document-understanding transformer, on a custom medical-billing dataset for structured field extraction, adding domain-specific tokens and resizing the decoder embeddings.
- Raised validation accuracy from 55% to 93% through improved preprocessing and model tuning.

Deep Learning Intern | **Tusk** Dec 2023 – Feb 2024

- Trained YOLOv8n on 1,133 curated, annotated images to 83% mAP@50 for toothbrush health classification, and exported it to ONNX and TFLite for mobile deployment.

Research Intern (Capstone) | **Bhabha Atomic Research Centre (BARC)** Jul 2023 – Apr 2024

- Raised stitching quality to a PSNR of 86 with a custom alpha-blended wavelet bank (sym6 + bior3.5), optimized homography stitching and concurrency, up from 65 with a bit-manipulation baseline.

Projects

Voice-Controlled Robotic Arm (COFFIN) — M.S. Capstone, RIT Apr 2026

- Integrated faster-whisper speech-to-text (1.0–1.4 s), Gemini intent classification (<500 ms) and TensorRT-optimized YOLO for visual servoing, accelerating detection 5.2× over eager PyTorch (56.7 → 10.9 ms, 17.6 → 92 FPS).
- Drove the system to 100% pick reliability (10/10 trials) at ˜11.5 s voice-to-pick by hardening it against false-positive detections, servo brown-out and stale-frame capture.

SLM Compression and Benchmarking for Edge Deployment

- Reduced GPU memory by ˜55% with 8-bit quantization (bitsandbytes LLM.int8()) on TinyLlama 1.1B, profiling peak VRAM and latency across prompt types with pynvml and psutil.
- Held accuracy within 1% across HellaSwag, GSM8K and ARC with only +0.09 perplexity: near-lossless compression for edge deployment.

Quantization-Aware Training for Integer-Only Inference May 2026

- Implemented quantization-aware training in PyTorch for a bias-free 256-64-10 MLP with straight-through-estimator fake quantization, per-layer symmetric int8 weights and uint8 activations; integer-only inference reaches 97.43% on the MNIST test set.
- Calibrated the hidden-activation scale with an EMA observer and froze it for the final 25% of training so the weights adapt to the deployed scale.

CUDA Kernel Optimization for VGG16 Inference

- Replaced VGG16's convolution layers with custom CUDA C++ kernels (memory coalescing, constant memory, preallocated buffers): 49.2 ms vs 73.3 ms for a naive GPU kernel and 77.5 ms on CPU, ˜35% faster with identical accuracy.

Education

M.S., Computer Engineering | Rochester Institute of Technology Aug 2024 – May 2026
Courses: Deep Learning, Machine Intelligence, High Performance Architecture, Robot Perception

B.E., Electronics & Telecommunication | University of Mumbai (F.C.R.I.T.) 2020 – 2024
Courses: Artificial Neural Networks & Fuzzy Logic, Image Processing